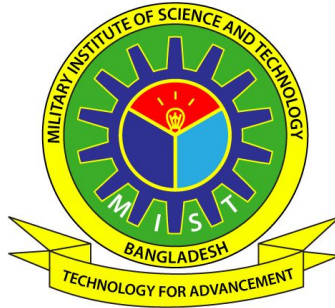


MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE-464: SOFTWARE DEVELOPMENT PROJECT-II

PROJECT REPORT

on

FoodRescue Sync

Commercial Surplus Food Logistics

Batch: CSE-23 **Section:** A **Group:** A6

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1 Problem Statement

Commercial food establishments generate waste massive amounts of food each day due to inefficiency and the absence of real-time logistics connecting them with local organizations or budget-conscious consumers. Surplus food that is still safe for consumption is discarded, contributing to lost revenue for businesses and land pollution. At the same time, a large portion of Bangladesh's urban population faces food insecurity, with no accessible way to benefit from the surplus available even close by.

The absence of a centralized, rapid-response platform means that potential food donors and recipients (who may be individuals or organizations) cannot coordinate efficiently. By the time food is identified as surplus, food begins to rot, making redistribution impractical. FoodRescue Sync addresses this gap by providing a real-time logistics platform that connects commercial food donors with organizations and consumers to report surpluses and ensuring food reaches those in need before it is wasted.

2 Project Objectives

1. To develop a circular economy and supply-chain logistics platform that eliminates commercial food waste by connecting donors with consumers in real time.
2. To provide businesses with revenue recovery streams and easier donation facility.
3. To combat food insecurity of the poor people by enabling organizations and individuals to claim donations and purchase surplus food at a discounted price.

3 Finalized Features

1. **Inventory Management:** Donors can upload and maintain inventory via barcode scanning or manual entry, enabling fast and accurate stock tracking.
2. **Expiry Tracking:** The system monitors each item's expiry date and automatically applies a "surplus" tag when the threshold is approached, triggering redistribution workflows.
3. **Flash Sales:** Near-expiry items are listed on a discounted marketplace, giving donors an opportunity to recover sunk costs while minimizing waste.
4. **Donation Logging:** Completed donations are automatically recorded, and receipts are generated and stored for the donor's records.
5. **Surplus Radar:** A real-time map displays active surplus food listings in the user's vicinity, en-

abling consumers to locate available food quickly.

6. **Bulk Claiming:** Organizations can instantly claim large batches of donated food through a bulk-claim workflow designed for high-volume pickups.
7. **Discount Marketplace:** A consumer-focused marketplace lists discounted surplus from nearby restaurants and caterers for direct purchase.
8. **Custom Alerts:** Push notifications alert users about new listings matching their preferred food types or favorite establishments.

4 Entity-Relationship (ER) Diagram



Figure 1: Entity-relationship diagram for FoodRescue Sync

5 Tentative UI Pages

Table 1: Tentative UI pages and their descriptions

Page	Description
Welcome Screen	Introduces the app and guides new users through initial role selection (donor, organization or consumer).
Login/Register	Allows users to sign in to an existing account or create a new one with role-based registration.
Donor Dashboard	Central hub for donors to view inventory summary, active entries and quick action shortcuts.
Add/Scan Inventory	Enables donors to add items via barcode scan or manual form with quantity and expiry date.
Expiry Tracker	Displays a sorted list of items approaching expiry with countdown timers.
Create Listing	Allows donors to list a surplus item as a flash sale or donation with price and pickup window.
Listing Image Manager	Provides donors with an interface to capture, upload, and preview food item photos during inventory listing.
Donation Log	Shows a history of completed donations with status and downloadable tax-deductible receipts.
Organization Dashboard	Central hub for organizations showing available donation listings, active requests, and upcoming pickups.
Organization Profile Setup	Enables non-profit entities to register and manage their institutional details, verification status, and contact information.
Surplus Radar (Map)	Real-time map of nearby active surplus food listings with type, distance, and category filters.
Bulk Request	Allows organizations to select multiple donation listings and submit a consolidated request in one step.
Request Status Tracker	Allows organizations and consumers to view the real-time approval status of their submitted food requests.
Pickup Coordination Screen	Displays pickup addresses, designated times, and logistical status updates for both donors and recipients.
Consumer Marketplace	Displays discounted surplus meals from nearby vendors with search and category filters.

Page	Description
Notification Centre	Shows personalised push notifications for new listings matching the user's saved preferences.
Profile & Settings	Allows users to update account details, food-type preferences, and alert notification settings.

6 Tentative APIs

6.1 Authentication

Table 2: Authentication API endpoints

Endpoint	Description
POST /auth/register	Register a new user with a chosen role (donor, organization, or consumer).
POST /auth/login	Authenticate a user and return a session token.
POST /auth/logout	Invalidate the current user's active session.

6.2 Food Listings

Table 3: Food listings management API endpoints

Endpoint	Description
POST /listings	Create a new food listing with item details, expiry date, and listing type.
GET /listings	Retrieve all active listings, filterable by type, category, or proximity.
GET /listings/:id	Fetch the full details of a single listing by ID.
PATCH /listings/:id	Update an existing listing's price, quantity, or status.
DELETE /listings/:id	Remove a listing once the food is no longer available.
GET /listings/nearby	Return active listings within a given radius of the user's coordinates.

6.3 Requests

Table 4: Requests API endpoints

Endpoint	Description
POST /requests	Create a new request for a specific food listing.
GET /requests/:id	Retrieve the specific details and current status of a request.
PATCH /requests/:id	Update a request's status (e.g., pending, accepted, or rejected).
GET /requests/user/:userId	Fetch all requests made by a specific user.

6.4 Pickup Assignments

Table 5: Pickup assignments API endpoints

Endpoint	Description
POST /pickups	Initialize a scheduled pickup tracking entry once a request is accepted.
GET /pickups/:id	Fetch logistics details, timing, and fulfillment tracking for a pickup.
PATCH /pickups/:id	Update the operational status of a pickup (e.g., scheduled, en-route, completed).

6.5 Food Images

Table 6: Food images API endpoints

Endpoint	Description
POST /listings/:id/images	Upload and link a photo asset to a specific food listing ID.
DELETE /images/:id	Remove an uploaded food image asset from the system.

6.6 Organizations

Table 7: Organizations API endpoints

Endpoint	Description
POST /organizations	Create an institutional organization profile associated with a user ID.
GET /organizations/:id	Retrieve profile details for a verified organization.

6.7 Donation Logs

Table 8: Donation logs API endpoints

Endpoint	Description
POST /donations	Log a completed donation and trigger automated tax receipt generation.
GET /donations/:userId	Retrieve all donation records and receipts for a given donor.

6.8 Notifications

Table 9: Notifications API endpoints

Endpoint	Description
GET /notifications/:userId	Fetch all notifications for a specific user, ordered by recency.
PATCH /notifications/:id	Mark a specific notification as read.

7 App Reviews

Each group member reviewed one existing food rescue or surplus food platform to identify best practices and some key gaps that FoodRescue Sync hopes to address.

7.1 Too Good To Go (Reviewed by: Lt Mashfiq Ahmed Chowdhury)

Too Good To Go [1] is a commercial surplus food marketplace operating in over 17 countries that connects consumers with restaurants and stores selling near-expiry food in discounted “Magic Bags.”

Table 10: Too Good To Go: features and limitations

Sl. No.	Features	Limitations
1	Magic Bag system: "bags" or packages of near-expiry food sold at a significant fixed discount	No organization or charity donation module. unsuitable for bulk redistribution
2	Location-based map showing partner restaurants and stores	Individual item selection not possible, only "bags" available
3	Secure in-app payment processing for consumers	Inventory management and expiry tracking not available to donors
4	Personal impact dashboard showing meals saved and carbon dioxide reduced	Not operational in Bangladesh
5	User ratings and reviews	No specialized push notification system

7.2 OLIO (Reviewed by: Lt Mehedi Hasan Redoy)

OLIO [2] is a UK-based community sharing platform for free food and household item exchange. Its Food Waste Heroes program enlists volunteers to collect unsold surplus food from supermarkets and redistribute it locally.

Table 11: OLIO: features and limitations

Sl. No.	Features	Limitations
1	Free food and item sharing between community members using posts with photo	No expiry tracking for businesses
2	Location based matching of listings	No automated donation receipt generation
3	Food Waste Heroes: volunteer driven supermarket extra food redistribution program	All transactions are free, no profiting
4	In-app direct messaging between listers and recipients	No bulk claiming feature
5	–	Platform dependent on volunteers

7.3 Food Rescue US (Reviewed by: Effat Jahan Erin)

Food Rescue US [3] is a US-based non-profit platform that matches food donors (restaurants, caterers, grocers) with registered non-profit recipient agencies. In doing this, the platform uses trained volunteers known as “Food Runners” for pickups.

Table 12: Food Rescue US: features and limitations

Sl. No.	Features	Limitations
1	Direct matching of verified food donors with registered non-profit organizations	Completely volunteer dependent
2	Real-time donation status tracking for donors, agencies, and volunteers	No inventory management, expiry tracking, or automated surplus-tagging tools
3	Total impact reporting in dashboard	No flash sale feature
4	Volunteer scheduling and management system	Only available in the United States

7.4 Flashfood (Reviewed by: Wasif Darain)

Flashfood [4] is a North American grocery surplus marketplace that partners with major supermarket chains to list near-expiry grocery items at discounts of up to 50%, available for in-store pickup.

Table 13: Flashfood: features and limitations

Sl. No.	Features	Limitations
1	Store pickup for groceries near expiry at 50% discount	Covers grocery stores only. Does not support restaurants, caterers and event venues
2	Wide partnership network in the US	Service only available in the United States
3	QR code-based pickup verification at the store counter	No real-time expiry tracking
4	Consumer savings tracker that shows savings	The platform is purely commercial
5	–	No alerting for listings. Users must browse to discover available items

In all, these apps show the viability of food waste reduction mobile apps and provide good user interfaces (UI). Across all platforms, critical features are missing, such as expiry tracking, inventory management

tools, bulk claiming for organizations and customized alert systems. In addition, some of these apps are purely volunteer driven and some are purely commercial and none are available in Bangladesh, so their action in practice could not be observed. FoodRescue Sync hopes to fill most of these gaps, especially in a Bangladeshi context.

8 Contribution Table

Table 14: Group member contributions

ID No.	Member Name	Contribution
202114191	Lt Mashfiq Ahmed Chowdhury	Problem Statement, App Review (Too Good To Go), Presentation Creation
202114196	Lt Mehedi Hasan Redoy	Idea of the Project, Project Objectives, Finalized Features, App Review (OLIO)
202314020	Effat Jahan Erin	ER Diagram, Tentative UI Pages, App Review (Food Rescue US)
202314053	Wasif Darain	Technology Stack Research, Report Compilation, Tentative APIs, App Review (Flashfood)

9 References

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